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Hi-CBM: Mitigating information leakage via hierarchical concept bottleneck modeling

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ABSTRACT

Concept Bottleneck Models (CBMs) enhance interpretability and facilitate effective intervention by explicitly mapping input features to labels through human-understandable concepts. However, existing CBM frameworks often suffer from *information leakage*, wherein latent unintended information bypasses the concept layer, undermining interpretability and contaminating downstream predictions. To address this challenge, we propose Hi-CBM, a refined CBM framework that explicitly safeguards the two inherent mappings in CBMs—*features* → *concepts* and *concepts* → *classes*—to prevent leakage. Specifically, a *Concept-Bottleneck Pooling* mechanism regulates the feature-to-concept mapping by selectively aggregating latent features into semantic concepts to filter out irrelevant signals, while a binary *Intervention Matrix* governs the concept-to-class mapping by constraining concept–class associations, preventing unintended information encoded in concept probabilities from influencing final predictions. Extensive experiments across multiple datasets show that Hi-CBM substantially mitigates information leakage and produces concept representations that are both interpretable and intervenable, while maintaining strong predictive performance.

1. Introduction

Deep neural networks (DNNs) have achieved remarkable success in many real-life tasks [1–3]. However, their black-box nature makes the extracted features obscure, hindering users from interpreting DNN predictions [4–8]. To address this issue, Explainable AI (XAI) has recently emerged with the purpose of delivering high-quality explanations for DNN predictions [9–13]. Concept Bottleneck Models [9,14–18] are a type of XAI models that make predictions based on human-understandable concepts. CBMs typically have a Concept-Bottleneck (CB) layer located before the last fully-connected (FC) layer. The CB layer takes features extracted (by preceding layers) from the input and maps them to a set of concepts. This effectively aligns the intermediate layers of a DNN with some pre-defined expert concepts, and the last FC layer determines the final label over those concepts. Often, CBMs require first training the CB layer to align each neuron to a concept that is pre-defined and understandable to humans.

Based on the output concepts of the CB layer, CBMs deliver two key benefits: *interpretability* and *intervenability*. First, users can interpret the predicted labels by inspecting the involved concepts. Second, users can intervene in the predicted labels by determining which concepts are involved in the prediction. Technically, CBMs are often categorized as soft CBMs and hard CBMs. The CB layer in soft CBMs outputs a probability (i.e., a number between 0 and 1) for each concept, whereas the hard CB layer outputs a binary state (i.e., 0 or 1) to indicate if a concept exists in the input. Then, a label predictor (often the last FC layer) predicts the final label based on the concept probabilities or states.

Despite the encouraging potential of CBMs in delivering human-understandable explanations, their interpretability and intervenability are largely undermined in de facto technical solutions due to *information leakage* [19–21], where unintended information beyond the concepts is leveraged by the label predictor. This issue is not merely anecdotal; empirical studies have shown that information leakage is prevalent across both synthetic and real-world datasets [19]. When inspecting predictions at the concept level, users cannot be certain whether a concept is activated because it truly appears in the input or because its representation carries other class-related signals. Such behavior directly undermines the interpretability and reliability that CBMs are designed to provide.

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Current approaches to mitigate information leakage in CBMs remain limited and fail to fully address the problem. The concept probabilities in soft CBMs may encode class distribution information, such that the label predictor can classify distinct class labels based on *indistinguishable* concepts [19]. For instance, the label predictor may leverage the probability differences of "`head`" and "`tail`" to classify dog vs. cat and achieve high accuracy, despite that these two concepts are insufficient to distinguish dog and cat. On the other hand, hard CBMs replace probabilities with binary indicators in an attempt to prevent leakage and were previously believed to be resilient to this problem [19]. However, a recent study Mahinpei et al. [20] has pointed out that hard CBMs can leverage unrelated hard concepts to convey class distributions to the label predictor. For example, Mahinpei et al. [20] demonstrate that hard CBM's performance can be improved by adding *meaningless* hard concepts.

To faithfully achieve CBM's design objectives, it is urgent to mitigate information leakage in CBMs. To this end, this paper proposes a novel paradigm of CBMs, dubbed as Hi-CBM, explicitly designed to mitigate information leakage and faithfully achieve CBM's design objectives. To this end, we first construct a carefully designed two-level hierarchy of concepts—*Perceptual* and *Descriptive*—advocating that *the concept set should consist of perceptual concepts that can be directly observed by humans without additional reasoning*. Building upon this hierarchical concept set, Hi-CBM mitigates information leakage through two complementary modules: (1) a hierarchical pooling mechanism at the concept bottleneck layer, selectively aggregating latent features into semantic concepts to effectively filter out irrelevant concepts; and (2) a binary *intervention matrix*, controlling concept-class associations and explicitly preventing unintended information encoded in concept probabilities from leaking into predictions. Collectively, these design enhancements ensure effective mitigation of information leakage.

Our comprehensive empirical evaluations across multiple diverse datasets demonstrate that Hi-CBM not only effectively mitigates information leakage (Section 4.4), but also consistently yields predictions that are more accurate (Section 4.3), interpretable (Section 4.5), and amenable to human intervention (Section 4.6), surpassing current state-of-the-art CBMs. These experimental findings illustrate that our proposed approach, Hi-CBM, delivers significant and measurable improvements, establishing a new benchmark in faithful and human-aligned Concept Bottleneck Models. Hi-CBM makes the following contributions:

- We develop a two-level set of concepts—*Perceptual* and *Descriptive*. Such a design can ensure perceptual-level concepts remain directly observable, thereby aligning model explanations closely with human intuition.
- Our design incorporates a hierarchical pooling mechanism and a sparse binary intervention matrix, ensuring concept predictions remain relevant and that no unintended predictive information can be leveraged to influence the final label predictions.
- Extensive empirical evaluations across diverse datasets demonstrate that the proposed approach effectively reduces information leakage and surpasses existing CBM methodologies in accuracy, interpretability, and human intervenability, establishing a new baseline for faithful and human-aligned CBMs.

2. Related works

The concept bottleneck paradigm has recently become one of the most influential frameworks in interpretable and interventional AI. It seeks to bridge the gap between human-understandable concepts and machine representations by introducing an intermediate, semantically meaningful layer between the input and the output. This section reviews the evolution of Concept Bottleneck Models (CBMs) and their post-hoc extensions, then discusses the key issue of *information leakage* that limits their interpretability and reliability. We finally position our proposed Hi-CBM within this context, highlighting how it differs from and improves upon existing approaches.

2.1. Concept bottleneck models

Concept Bottleneck Models (CBMs) [9,14–18] are a type of XAI models that incorporate an intermediate conceptual representation layer connecting raw input features to the final model outputs. Specifically, given an input image $x^{(i)} \in \mathbb{R}^d$, a CBM first maps this image into a conceptual feature vector $C^{(i)} \in [0,1]^k$, where k is the total number of predefined concepts. Each component $C_j^{(i)}$ of this vector encodes the relevance or presence of the j-th concept in the given image. Depending on how these conceptual representations are formulated, CBMs typically fall into two variants: the Soft CBM, where each concept value $C_j^{(i)}$ is a continuous probability representing the degree of concept presence, and the Hard CBM, where each $C_j^{(i)}$ is strictly binary, indicating the absence or presence of the concept.

In practice, CBMs require supervision of their intermediate concept representations. Given a conventional dataset $D = \{(x^{(i)}, y^{(i)})\}_{i=1}^N$ consisting of N pairs of input images $x^{(i)} \in \mathbb{R}^d$ and corresponding ground truth labels $y^{(i)} \in \mathbb{N}$, CBMs necessitate the augmentation of this dataset with concept-level annotations. Specifically, for a predefined set of concepts C, each image $x^{(i)}$ is annotated by a binary concept vector $c^{(i)} \in \{0,1\}^{|C|}$, where the j-th entry $c_j^{(i)} = 1$ indicates the presence of the j-th concept in image $x^{(i)}$, and a value of 0 indicates its absence. Obtaining these detailed concept annotations, however, typically involves considerable human effort, as annotators must manually inspect each image to determine the presence or absence of every predefined concept. This intensive annotation process significantly increases the labeling cost and limits the practical scalability as well as the broad applicability of CBMs in real-world settings.

2.2. Post-hoc concept bottleneck models

To mitigate potential challenges related to extendability and reliance on manual annotations in traditional CBMs, recent research has explored post-hoc CBMs [14–18]. Unlike conventional CBMs that require retraining, post-hoc CBMs utilize a pretrained neural network backbone and incorporate a flexible, soft, probabilistic concept bottleneck layer. This approach also enables automatic generation of interpretable concept annotations, providing post-hoc explanations while reducing the need for manual labeling.

We categorize related post-hoc CBM works into three main groups.

(1) Knowledge graph-based methods. Representative work such as [14] utilizes general-purpose knowledge graphs (e.g., ConceptNet [22]) to extract automatic concept supervision. Although these approaches benefit from structured knowledge, the concepts are fixed and often limited in number, making it difficult to capture nuanced, domain-specific attributes.

(2) Large language model (LLM)-based methods. With the advent of powerful LLMs [23], subsequent works [15–18] leverage them to automatically generate discriminative concept annotations. For example, [15] transforms black-box neural networks into interpretable CBMs without labeled concepts, achieving comparable accuracy; [16] proposes LaBo, which uses GPT-3 to generate factual candidate concepts and selects diverse bottleneck representations via a submodular utility function; [17] matches visual embeddings of detected parts to textual descriptors derived from class names; and [18] searches for compact, high-utility attribute subsets from LLM-generated candidates.

(3) Vision-language model (VLM)-based methods. The most recent line of work explores leveraging VLMs to directly couple visual and textual concept spaces [24–26]. [24] is among the first to employ a VLM to select textual concepts that describe the target dataset classes; [25] builds upon this idea by constructing a Vision-to-Concept (V2C) tokenizer that explicitly quantizes images into visual concepts; and [26] applies VLM-based bottleneck concepts to medical imaging, such as skin lesion classification.

2.3. Information leakage in CBMs

As studied in prior works [19–21], the interpretability and intervenability, two key design objectives of CBMs, are undermined by the information leakage issue:

Definition 2.1 (Information Leakage). *Information leakage in CBMs indicates that unintended and additional information beyond the predicted concepts is leveraged for the follow-up label prediction.*

Leakage in hard CBMs. While hard CBMs were previously believed to be resilient to information leakage, one recent work pointed out that hard CBMs can encode/leak additional information via irrelevant (hard) concepts [20]. As demonstrated in [20], merely adding *meaningless* hard concepts to hard CBMs can improve their label prediction performance, indicating that these added concepts bring additional information to the label prediction.

Leakage in soft CBMs. In soft CBMs, the information leakage is induced by the concept probabilities [19]. Consider a CBM-based animal classifier with the concept set {`fur`, `tail`}, suppose it predicts `fur` with a higher probability than `tail` when recognizing dogs, but assigns `tail` a higher probability than `fur` when identifying cats. Although the concepts `fur` and `tail` are insufficient to distinguish dog vs. cat, the label predictor (which is also a black-box DNN) may compare concept probabilities to decide the final label. Apparently, the CBM-based classifier does *not* only rely on `fur` and `tail` to classify cat and dog, and the interpretation and intervention based on `fur` and `tail` are consequently invalid. Here, the difference between concept probabilities encodes the class distribution information that is beyond the concepts themselves.

Essentially, the information leakage is primarily induced by ⓐ *irrelevant concepts* and/or ⓑ *additional information encoded in concept probabilities*. Hi-CBM eliminates these two leakage sources by deliberately reforming the adoption of concepts in the label prediction, as will be introduced in Section 3. Hi-CBM also achieves post-hoc CBM conversion by only training a light-weight one-layer CB layer for each backbone model.

2.4. Our method in context

Building upon the preceding discussion of existing post-hoc CBMs and their susceptibility to information leakage, we now position our approach in relation to prior work. Knowledge graph-based methods [14] and large language model-based approaches [15–18,23] often face challenges in achieving broad coverage and strong domain relevance. In contrast, our approach structures concepts hierarchically rather than relying on a single flat predefined set, enabling richer semantic coverage and closer alignment with target tasks. Recent vision–language model techniques [24–26] capture visual–text associations to construct concept spaces, but can suffer from hallucinations that undermine reliability. To counter these tendencies, our design employs concept–class aware training strategies that enhance concept quality and maintain alignment with downstream tasks; to address the information leakage present in all previously discussed approaches, it further integrates dual-level mechanisms for robust mitigation. The specific designs will be detailed in Section 3, with empirical validation in Section 4.

3. Hi-CBM's design

In this section, we present the overall workflow of Hi-CBM, illustrated in Fig. 1. The workflow consists of three main stages: (i) *Concept Set Construction* (Section 3.1), (ii) *Preventing Information Leakage with Dual-Level Mechanisms* (Section 3.2), which encompasses the two core components — *Concept-Bottleneck Pooling* (Section 3.2.1) and *Label Prediction via Intervention Matrix* (Section 3.2.2) — together with an analytical justification of how Hi-CBM mitigates label leakage (Section 3.3), and (iii) *Training Objectives* (Section 3.4).

3.1. Concept set construction

Hierarchical Concepts. In Hi-CBM, the concept set is organized as a two-level hierarchy to better capture the semantics relevant to label prediction while maintaining interpretability. Formally, the entire concept vocabulary can be represented as a collection of paired elements $C = \{\langle \{\text{CEP}\}_i, \{\text{DESCR}\}_i \rangle\}_{i=1}^{p\times q}$, where p denotes the number of *Perceptual Concepts* (nounal visual parts) considered for each class, and q denotes the number of *Descriptive Concepts* (adjectival attributes) associated with each perceptual concept. Each $\langle \{\text{CEP}\}_i, \{\text{DESCR}\}_i \rangle$ pair consists of a first-level perceptual concept $\{\text{CEP}\}_i$ (the visual part or component) and its corresponding second-level descriptive concept DESCR_i (its attribute). Each pair is further associated with one or more class labels that indicate its relevance in prediction.

The first level consists of *Perceptual Concepts*, defined as clear, nounal visual parts or attributes that can be directly observed from the image without prior domain reasoning. Examples include `wing`, `eye`, `tail`, `leaf`, or `hand`. These concepts correspond to tangible, visually discernible elements that both humans and models can consistently recognize. Formally, a perceptual concept is a noun referring to a distinct and observable component or attribute in the image, such that its presence can be verified without background knowledge.

The second level consists of *Descriptive Concepts*, which refine the perceptual concepts by specifying their salient properties along orthogonal dimensions such as shape, color, or size. Examples include `red wing`, `big eye`, `long tail`, or `round leaf`. Formally, a descriptive concept is an adjective or short phrase that modifies a perceptual concept to indicate a distinguishing feature, drawn from predefined dimensions (shape, color, size) to minimize ambiguity.

This hierarchical structure enables Hi-CBM to indicate not only *which* perceptual concepts are involved in the prediction (coarse-level interpretation) but also *how* those concepts appear in the given input (fine-grained interpretation). For example, in classifying different breeds of

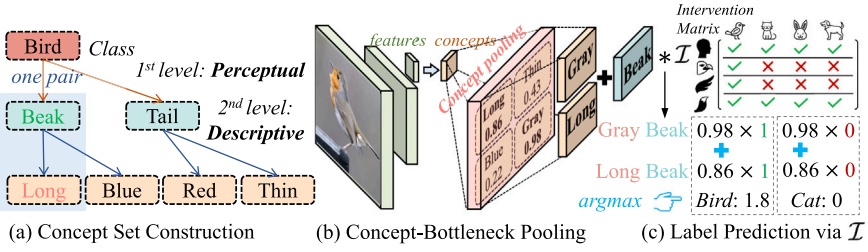


Fig. 1. Workflow of Hi-CBM. First, during concept set construction (a), we annotate a comprehensive set of concepts and organize them into a two-level hierarchy—*Perceptual* and *Descriptive*. Next, we train a Concept-Bottleneck layer with Pooling mechanism to selectively map features extracted by the backbone model onto these concepts (b). Finally, we multiply the concept scores obtained from the CB layer by an intervention matrix $\mathcal{I}$, which encodes binary class-to-concept relationships, to predict class labels (c).

fish in CIFAR-100, the nounal perceptual concept `tail` might be present for all fish classes, while its descriptive attributes—such as `fan tail` for `goldfish` or `veil tail` for `betta fish`—provide discriminative details vital for label prediction.

Prompt Design. We use GPT-4 [27] to automatically build the concept set. Since we require perceptually perceivable concepts to enhance CBM interpretability and intervenability, we first query the GPT model with the following prompt:

> "*To identify {CLS} visually, please list the most important {p}* ***visual*** *parts which a {CLS} has.*"

Here, the token {CLS} can be replaced by any classes the original feature-based model can predict. The GPT returns a set of nounal visual parts in {CLS} as our first-level concept {CEP}. Then, for each concept {CEP} of a class {CLS}, we query the GPT model with the following prompt to uncover semantics associated with different concepts.

> "*To visually identify {CLS}, please describe the {q} most common characteristics of {CLS}'s {CEP} from the three dimensions of shape, color, or size.*"

With the above prompt, the GPT model can return to us a set of adjectival concepts as our second-level concepts that describe the semantics of the first-level nounal concepts. We limit the adjectival concepts to follow the three orthogonal dimensions of shape, color, and size, thereby reducing ambiguity. We also remove any concept that is longer than 40 characters to make the descriptions simple. This way, we obtain a two-level concept set containing concepts represented in a $\langle\{\text{CEP}\},\{\text{DESCR}\}\rangle$ form, as illustrated in Fig. 1, e.g., $\langle \texttt{Beak}, \texttt{Long}\rangle$ constitutes one $\langle\{\text{CEP}\},\{\text{DESCR}\}\rangle$ pair for the input `bird` image. Accordingly, we also know each $\langle\{\text{CEP}\},\{\text{DESCR}\}\rangle$'s relevant class {CLS}.

Extendability. Our concept set is highly extendable. To convert backbone models whose supported classes are not included in our concept set, users can simply replace the {CLS} in the first prompt and query GPT models to obtain the corresponding concepts, and accordingly update the {CLS} and {CEP} in the second prompt to augment the concept set with new $\langle\{\text{CEP}\},\{\text{DESCR}\}\rangle$ pairs.

3.2. Preventing information leakage with dual-level mechanisms

To mitigate the pervasive problem of information leakage in CBMs, Hi-CBM introduces a dual-level prevention framework. At the first level, the model applies a pooling mechanism at the concept bottleneck to automatically filter out irrelevant or spurious concepts before classification. At the second level, an intervention matrix enforces an explicit and interpretable mapping between concepts and classes, ensuring that only intended concepts contribute to label prediction. Together, these mechanisms suppress leakage from both irrelevant concept activations and uncontrolled concept–class associations, leading to more robust and transparent predictions.

3.2.1. Concept-bottleneck pooling

A key difference between Hi-CBM and previous CBMs lies in the structure of the Concept-Bottleneck layer. As illustrated in Fig. 1(b), Hi-CBM incorporates a pooling mechanism at the concept bottleneck layer, selectively aggregating latent features into semantic concepts to filter out irrelevant concepts for each image x.

This "pooling" layer is analogous to a one-dimensional hierarchical max pooling operation [2]—commonly used in CNNs to retain the most salient activations while discarding less relevant ones—applied here across conceptual dimensions rather than spatial dimensions. Formally, let $c_{i,j}$ denote the activation score of the i-th perceptual concept and its j-th descriptive sub-concept. For a pooling kernel of size p (over perceptual concepts) and stride q (over descriptive concepts), the pooled output at position (u, v) is defined as

$$m_{u,v} = \max_{(i,j)\in\mathcal{R}_{u,v}} c_{i,j}, \tag{1}$$

where $\mathcal{R}_{u,v}$ denotes the set of concept activations within the receptive field centered at (u, v), determined by p and q. This operation preserves the most semantically relevant concept in each local region, effectively compressing information and filtering out noise. Analogous to max pooling in CNNs, it reduces redundancy and mitigates over-reliance on spurious features.

The pooling operation in Hi-CBM thus "fuses" concept predictions at multiple hierarchical levels, aiming to remove irrelevant concepts before classification. As a result, it not only enhances the performance of CBMs but also reduces potential information leakage from unrelated concepts (a).

Specifically, we first compute the similarity between the input image x and each concept $\langle\{\text{CEP}\},\{\text{DESCR}\}\rangle$ involved in predicting the class label y. Similarity is computed via cosine similarity between the CLIP [28] embeddings of x and each concept pair $\langle\{\text{CEP}\},\{\text{DESCR}\}\rangle$. For each {CEP}, we define the *number of retrieved concepts* k, selecting the top-k concepts by maximizing the highest similarity scores as the final concept predictions. Consequently, each input image is associated with a total of $p \times k$ predicted concept annotations.

3.2.2. Label prediction via intervention matrix

To explicitly control how concepts contribute to label prediction and to further eliminate any residual information leakage, we introduce an *intervention matrix*. This module serves as a transparent and controllable interface between the concept bottleneck layer and the final classification head. Intuitively, it determines which concepts are allowed to influence each class decision, enforcing a precise concept-to-class correspondence. By defining these relationships in a binary and interpretable manner, the intervention matrix not only prevents unwanted information propagation from soft concept activations but also enables straightforward human intervention—allowing users to directly adjust the model's reasoning process by adding or removing specific concepts.

The predicted label is decided as $\arg\max c * \mathcal{I}$, as shown in Fig. 1(c), where $\mathcal{I}$ is a intervention matrix, and $*$ denotes matrix multiplication. That is, the j-th class's prediction confidence l is computed as

$$l_j = \sum_{i=1}^{p\times q} c_i * \mathcal{I}_{i,j} \tag{2}$$

Here, to further mitigate potential information leakage from the soft concepts predicted by the CB layer (b), we introduce an intervention matrix $\mathcal{I}$, which indicates in a binary manner the concepts (i.e., pairs of $\langle\{\text{CEP}\},\{\text{DESCR}\}\rangle$) that should be considered during prediction. Specifically, $\mathcal{I}$ is a binary matrix of shape #concepts × #classes. Each $\mathcal{I}_{i,j} \in \{0, 1\}$ and $\mathcal{I}_{i,j} = 1$ indicates that the i-th concept should be involved in predicting the j-th class; otherwise, $\mathcal{I}_{i,j} = 0$. The $\mathcal{I}$ is obtained after querying the GPT with the second prompt mentioned in Section 3.1.

Intervenability. Supported by the intervention matrix, Hi-CBM provides improved intervenability through the intervention process [15], which removes incorrectly-predicted concepts to check if the prediction changes to the ground-truth label class. Essentially, this process and its objective are consistent with our intervention matrix where a binary matrix is employed to decide which concepts are used for predicting labels. The intervention matrix in Hi-CBM is a binary matrix where each (i, j)-th entry indicates whether the i-th concept is set as involved in predicting the j-th class. The intervention matrix also does not have any trainable parameters; its binary entries are set up by users. To perform intervention for an input x, users only need to accordingly update entries in the intervention matrix. For instance, setting the (i, j)-th entry as 1 allows the i-th concept to be involved in predicting the j-th label for x. When ground truth concepts for the j-th class are available, to intervene in the label prediction of Hi-CBM accordingly, users only need to set the corresponding entries (of the ground truth concepts) in the intervention matrix's j-th column as 1 and let other entries in the j-th column be 0.

3.3. Analytical justification of label leakage elimination in hi-CBM

Hi-CBM eliminates information leakage from both sources mentioned in Section 2. First, for leakage due to irrelevant concepts (ⓐ), we mitigate it during the CB layer: Hi-CBM only focuses on concepts that should be involved in predicting the label. Second, we further select those most important descriptive concepts {DESCR} via our concept pooling mechanism, thus trimming irrelevant concepts. With these two steps, irrelevant concepts are rarely kept in label predictions. Moreover, for leakage where the concept probabilities encode distribution of class information (ⓑ), we justify how our intervention matrix can eliminate such leakage. Consider two classes a and b whose involved concepts (according to $\mathcal{I}$) constitute sets C_a and C_b, respectively. We define $S = C_a \cap C_b$, $A = C_a \setminus S$, and $B = C_b \setminus S$. That is, S consists of concepts shared by both classes, and A and B denote the sets of concepts that are only involved in predicting a and b, respectively.

① If $S = \emptyset$, i.e., the two classes do not share any concepts, it should be clear that classifying a and b only relies on their disjoint concepts A and B, and Hi-CBM finds concepts that uniquely exist in classes a and b. Note that our concept pooling mentioned above can help reduce the size of S, increasing the chance of $S = \emptyset$.

And in case $S \neq \emptyset$, i.e., the two classes share some concepts, we analyze the following two cases.

② If $A = B = \emptyset$, i.e., predicting class a and b rely on the same *indistinguishable* concepts. Since our label prediction confidence (i.e., $c * \mathcal{I}$) is the sum of concept probabilities, it cannot distinguish a and b, and no information beyond the concept is leveraged to falsely conduct classification.

③ If $A \neq \emptyset$ or $B \neq \emptyset$, i.e., either class a or b, or both, have their unique concepts. Thus, given an input, the difference between the two label prediction confidences (w.r.t. class a and b) is only induced by the concepts uniquely belonging to A and B. Apparently, no information beyond a and b's unique concepts is leveraged to classify them.

3.4. Concept-class aware training objectives

Unlike previous post-hoc CBMs that directly train the follow-up label predictor over concept similarities (which undermines the intervenability as pointed out by [29]), we do not train a label predictor. Instead, we train the CB layer (without training the backbone model) since label supervision is unavailable for test images. Specifically, for each input image, we set the selected $p \times k$ concepts as hard labels and form $p \times k$ binary classification tasks following the conventional training procedure. These binary classification tasks are optimized with binary cross-entropy loss $\ell_{BCE}(c, GT_c)$, where c is the concept probabilities predicted by the CB layer and GT_c is the ground truth concepts. Both c and GT_c are $p \times q$-dimension vectors. Since $\mathcal{I}_{i,j}$ indicates whether the i-th concept should be involved in predicting the j-th class, the label j's confidence l_j equals the sum of probabilities of those involved concepts. Consistent with conventional CBM training, we have a cross-entropy loss $\ell_{CE}(l, GT_l)$, where l is a $\#classes$-dimensional vector and GT_l is an integer indicating the ground truth class label. By explicitly linking each concept to its related classes during training, the model maintains a grounded correspondence between visual evidence and label semantics. This targeted supervision effectively constrains the model's generative freedom, thereby mitigating the common hallucination issues often observed in large language or vision–language models.

It's worth noting that the intervention matrix $\mathcal{I}$ is always fixed (unless users intervene in Hi-CBM), and $\ell_{CE}(l, GT_l)$ is only leveraged to optimize the CB layer. Therefore, the overall training objective is to minimize:

$$\ell_{BCE}(c, GT_c) + \ell_{CE}(c * \mathcal{I}, GT_l). \tag{3}$$

Post-hoc Concept Mapper. It's worth noting that Hi-CBM also achieves post-hoc CBMs. Similar to existing post-hoc CBMs, to convert a new backbone model, Hi-CBM only requires training a light-weight layer. Differently, previous works train the label predictor using their pre-built concept set, while Hi-CBM only trains a one-layer CB layer without training any label predictor.

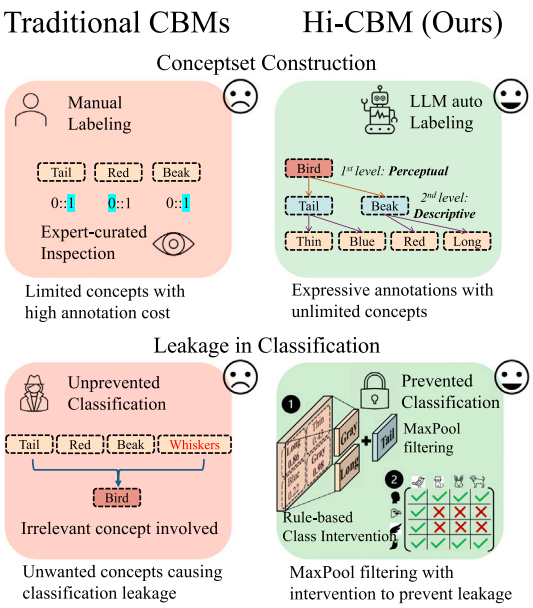


Fig. 2. Comparison between conventional CBM components and the proposed Hi-CBM counterparts.

3.5. Hi-CBM workflow and improvements over conventional CBMs

As illustrated in Fig. 1, our proposed Hi-CBM workflow comprises multiple components. To further clarify the key differences from traditional CBMs, Fig. 2 decomposes each component of Hi-CBM and compares it with its conventional counterpart.

Regarding the aspect of concept set construction, most existing CBMs rely on manual labeling. This approach suffers from several limitations: due to high annotation costs, it is impractical to scale up, leading to binary concept selection as shown in the figure. Consequently, only a small number of concepts can be labeled, and the process is time-consuming and non-automated. In contrast, our Hi-CBM leverages GPT-4 to perform fully automated, two-layer annotation, resulting in a highly expressive concept set. Moreover, the automation allows the generation of an unlimited number of concepts without human intervention.

With respect to class label prediction, traditional CBMs do not actively prevent irrelevant concepts from influencing the classification outcome. As a result, some unrelated concepts, such as "whiskers" (typically referring to canine facial hair) in a bird classification task, may inadvertently assist the model via unintended shortcuts, a phenomenon regarded as information leakage. To address this issue, Hi-CBM adopts a two-stage prevention strategy. First, MaxPooling is used to filter out irrelevant concepts before classification. Second, we design a rule-based intervention matrix for the final prediction stage, which explicitly enforces that only a fixed set of relevant concepts is allowed to contribute to the prediction of each class. This strict constraint effectively prevents unwanted information from leaking into the decision-making process.

4. Evaluations

We first introduce our experimental setup in Section 4.1, followed by an ablation study in Section 4.2. Next, we evaluate the label prediction performance of Hi-CBM in Section 4.3. We then assess information leakage from three concept-wise perspectives: (1) Removal, (2) Locality, and (3) Perturbation, detailed in Section 4.4. Finally, we evaluate

the interpretability and intervenability of Hi-CBM through a series of experiments presented in Sections 4.5 and 4.6 respectively.

4.1. Experimental setup

Datasets. Following previous works [14–16], we consider the following datasets to evaluate the performance of different models. These datasets are representative and collectively span three major computer vision domains, allowing us to examine model behavior in both general-purpose and domain-specific contexts. For general classification tasks, we adopt *CIFAR10* and *CIFAR100* [30], which provide low-resolution natural images over 10 and 100 object categories, respectively. Their popularity as standard benchmarks enables comparison with a broad body of prior work and ensures that our evaluation covers everyday visual concepts that are familiar to most models. For fine-grained, object-specialized classification, we select *CUB-Bird* [31], a dataset containing 200 bird species annotated with detailed part-level information. This domain requires distinguishing subtle visual differences between classes, providing a rigorous test of a model's capacity for precise and interpretable concept recognition. For medical image analysis, we use *HAM10000* [32], a large dermatoscopic image dataset that covers diverse pigmented skin lesions. This dataset represents a safety-critical application where domain-specific concept annotation is vital, and where interpretability and error prevention are especially important for clinical trustworthiness. Together, this selection enables a systematic and comprehensive assessment of CBM interpretability, intervenability, and robustness across domains of varying granularity and impact.

Prompt Design and Concept Generation. Hi-CBM employs GPT-4 (2023-07-01) [27] to generate (textual) concepts. Our explorations indicate that minor adjustments to the general prompt structure (e.g., wording, grammar, phrasing) have minimal impact on the generated concepts. This is primarily due to the straightforward nature of our prompts, which simply require generating clear descriptions for concepts and classes. However, we observed that careful selection of hyperparameters is essential when generating concept sets. While it may seem beneficial to generate a large set of concepts for completeness, overly large concept sets often introduce redundant or irrelevant information, increasing inference costs without meaningful benefit. Based on these insights, we selected a default configuration with stride $p = 5$, kernel $q = 6$, and retrieval size $k = 2$. The rationale behind these parameter choices is further detailed in Section 4.2.

CBP Layer & Training. The CBP layer in Hi-CBM is a single-layer fully-connected neural network. We train the CBP layer using the Adam optimizer with 100 epochs, which takes approximately 30 minutes. We consider learning rates of 0.00001, 0.0001, 0.001, and 0.002. All experiments are performed on an AMD Ryzen 3970X CPU with 256GB RAM and one Nvidia GeForce RTX 3090 GPU.

Backbone Selections. We carefully select backbone models for each dataset following established practices in the literature to ensure fair comparisons and optimal performance. Specifically, we mainly leverage the *CLIP* (Contrastive Language-Image Pretraining) model [28], a powerful vision-language pretrained framework trained on a large-scale dataset of image-text pairs. CLIP learns visual representations by aligning images and their corresponding textual descriptions in a shared embedding space, enabling strong zero-shot and transfer learning capabilities across diverse downstream tasks.

For CIFAR10 and CIFAR100, we adopt the *ResNet-50* variant of CLIP (*CLIP-RN50*), which integrates the widely-used *ResNet-50* convolutional architecture [3] as its visual encoder. This choice aligns with previous studies [14–16], which have demonstrated that CLIP-RN50 provides robust and reliable performance on general image classification benchmarks. For the CUB-Bird dataset, we utilize the *Vision Transformer* variant of CLIP (*CLIP-ViT14*) [28], following the recommendation of [16]. CLIP-ViT14 employs a transformer-based visual encoder architecture [33], which has been shown to excel in fine-grained visual recognition tasks due to its ability to capture subtle visual details and long-range dependencies within images. Lastly, for the HAM10000 dataset, we select the *Inception* architecture [34], pretrained specifically on dermatological images from HAM10000 [35]. This Inception backbone has been widely validated and adopted in dermatology-related image classification tasks, providing domain-specific representations that significantly enhance classification performance.

Overall, this backbone selection strategy ensures that each dataset benefits from a model architecture and pretrained weights that are both well-established and optimized for their respective visual recognition tasks (Table 1).

Table 1
Summary of datasets used in experiments.

Dataset Name	Class Type	Number of Classes	Training Samples	Validation Samples
CIFAR-10 [30]	General objects	10	50,000	10,000
CIFAR-100 [30]	General objects	100	50,000	10,000
CUB-Bird [31]	Bird species	200	5994	5794
HAM10000 [32]	Skin lesions	7	7000	3015

4.2. Ablation studies

The choice of kernel size q and stride p. In particular, we evaluated the number of unique textual concepts generated by GPT when using different p and q. We deem a textual concept as valid if 1) it is not generated by GPT twice, and 2) its synonyms are also not in GPT's outputs. We used CIFAR10 to speed up the experiment and fixed the prompt and GPT's temperature. Results for parameters p and q are shown in Tables 2 and 3, respectively. While all textual concepts are valid at $p = 2$, coverage improves with higher values and saturates around $p = 5$. Similarly, stride q saturates around $q = 6$. Thus, we set $p = 5$ and $q = 6$ by default for all subsequent experiments in this paper. This choice balances the trade-off between retaining meaningful concept information and avoiding redundant or irrelevant outputs. Specifically, lower threshold values would discard valuable concepts, whereas higher thresholds would lead GPT to produce repetitive or uninformative concepts.

Choosing the number of retrieved concepts k. Unlike the parameters p and q, which are fixed during the GPT-based concept generation phase, the parameter k requires more careful consideration, as it directly determines the final concept predictions per image. Since selecting more concepts (larger k) may inadvertently introduce irrelevant or redundant concepts, it is crucial to analyze potential information leakage caused by varying k. Therefore, we conduct an analysis of information leakage to better understand the impact of different k values. Among the three leakage testing approaches introduced in Section 4.4, we select the Concept Leakage Evaluation (CLE) method (described in Section 4.4.3) due to its intuitive nature and suitability for clear ablation studies. Specifically, the CLE allows us to observe the step-by-step trend in the leakage as k varies. The results of this analysis are illustrated in Fig. 3, and the corresponding Area Under Curve (AUC) values for each curve are summarized in Table 4.

Our findings indicate that varying k within a moderate range does not significantly affect information leakage, demonstrating the robustness and stability of our concept pooling mechanism with respect to this hyperparameter. Note that the concept pooling mechanism primarily aims to rule out irrelevant concepts to reduce hard-concept leakage, this result can justify that our concept set is condensed and unlikely to have redundancy. Considering that more predicted concepts ease the interpretation (as suggested in [16]), and to reduce the training complexity, we recommend $k = 2$, as adopted in the main body of this paper. Having that said, we suggest practitioners to adjust k based on their specific applications.

GPT Versions used in Concept Set Construction. Hi-CBM employs GPT-4 (2023-07-01) to generate (textual) concepts. To justify that the superiority of Hi-CBM is *not* brought by GPT-4, we also adopt

Table 2
Ablation study of p.

	Airplane	Automobile	Bird	Cat	Deer	Dog	Frog	Horse	Ship	Truck	Avg
$p = 2$	2	2	2	2	2	2	2	2	2	2	2
$p = 5$	5	4	5	5	5	5	5	5	5	4	4.8
$p = 10$	7	4	7	4	6	7	5	5	4	4	5.3
$p = 20$	7	5	6	7	5	7	4	5	5	4	5.5

Table 3
Ablation study of q.

	Wing	Tail	Door	Wheel	Headlight	Avg
$q = 3$	3	3	3	3	3	3
$q = 6$	6	6	5	6	6	5.8
$q = 9$	7	6	5	7	6	6.2

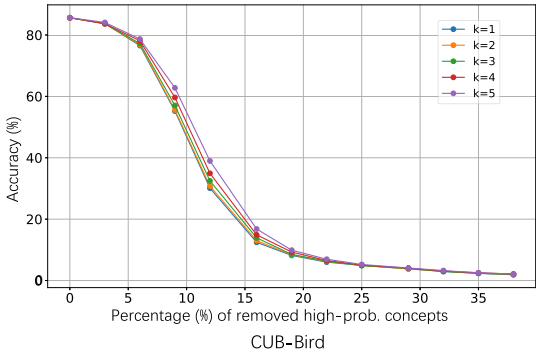


Fig. 3. Impact of the concept pooling parameter k on information leakage, evaluated using the Concept Leakage Evaluation (CLE) method on the CUB-Bird dataset. The figure illustrates that varying k within a moderate range leads to minimal changes in leakage.

Table 4
Area Under Curve results of Fig. 3.

Number of K	1	2	3	4	5
CUB-Bird	26.73	26.94	27.34	28.06	29.05

Table 5
Performance comparison of Hi-CBM with different GPT versions (GPT3 and GPT4).

Model	CUB-Bird	CIFAR10	CIFAR100	HAM10000
Hi-CBM + GPT4	**86.00**$_{\pm 0.04}$	88.97$_{\pm 0.18}$	**69.79**$_{\pm 0.23}$	83.69$_{\pm 0.30}$
Hi-CBM +GPT3	85.42$_{\pm 0.10}$	**89.21**$_{\pm 0.18}$	68.32$_{\pm 0.11}$	**83.85**$_{\pm 0.02}$

GPT-3 (2023-03-15) to generate concepts for Hi-CBM. Overall, Hi-CBM's performance on GPT-4 and GPT-3 is almost identical and constantly outperforms SOTA CBMs to a large extent. To justify that Hi-CBM's superiority over prior GPT-3-based CBMs is not due to GPT-4, this section evaluates Hi-CBM's performance with GPT-3. We use the same hyperparameters as in Section 4.1, namely $k = 2$, $p = 5$ and $q = 6$. Results are shown in Table 5 below: Hi-CBM's constantly outperforms prior SOTA CBMs when using different GPT versions, rendering the superiority of Hi-CBM's design and our novel mechanisms in CBMs. Different GPT versions may lead to (slightly) different performances on different datasets, though the overall performance of Hi-CBM is stable across different GPT versions.

General vs. Domain-Adapted GPT Concept. We assess whether general-purpose LLMs like GPT-4, as used in our Hi-CBM CBM, are qualified to generate concepts comparable to those produced by a domain-adapted LLM model. To this end, we compare Hi-CBM-generated concepts with those from a domain-adapted GPT on HAM10000 [32], as shown in Table 7. Specifically, we used Sentence-PubMedBert to compare the concept descriptions of seven HAM10000 label classes [32], created by medical experts, with textual knowledge generated by GPT-4 and MedGPT [36], a fine-tuned ChatGPT for the medical field, and calculated the cosine similarity score to evaluate the semantic alignment between the generated knowledge and expert descriptions. The results reveal that the concepts generated by GPT-4 align with the medical descriptions achieving semantic scores of at least 0.88 per class. MedGPT performed even better, with scores exceeding 0.93 for all classes. These results indicate that both GPT-4 and MedGPT are capable of generating high-quality medical concepts.

4.3. Label prediction evaluation

Table 6 reports the label prediction performance of all SOTA CBMs, the vanilla feature representation (Feat), and Hi-CBM across the four datasets. As shown, Hi-CBM consistently outperforms all existing post-hoc CBMs. For general-domain datasets such as CIFAR10 and CIFAR100, Hi-CBM achieves clear performance advantages over all baselines, even surpassing the feature-based upper bound (**Feat**) on CIFAR10 with an accuracy of 88.97%. We attribute these results to the use of structured, dual-level concept representations in Hi-CBM, which capture richer semantic information and thereby overcome the representational limitations of traditional CBMs. In addition, the refined pooling layer and rule-based intervention matrix together help suppress unwanted information propagation, further boosting model accuracy.

For specialized datasets, including the fine-grained CUB-Bird and the medical HAM10000 benchmarks, the evaluation becomes considerably more challenging. On CUB-Bird, most baselines exhibit a noticeable performance drop—typically more than five percentage points below

Table 6
Performance comparison with SOTA CBMs and the vanilla setting **Feat**. We mark the best performance in **bold**. "N/A" indicates that they do not support the related concept generation. For each experimental setting, we perform four runs in total, using learning rates of 0.00001, 0.0001, 0.001, and 0.002. We report the mean and standard deviation of the results obtained across these runs.

Model	CUB-Bird	CIFAR10	CIFAR100	HAM10000
Feat	86.41	88.80	70.10	84.07
P-CBM	78.18$_{\pm 0.23}$	81.23$_{\pm 0.22}$	60.00$_{\pm 0.01}$	72.37$_{\pm 0.21}$
Label-Free	78.84$_{\pm 0.10}$	85.50$_{\pm 0.64}$	65.19$_{\pm 0.06}$	81.78$_{\pm 0.18}$
Labo	83.22$_{\pm 0.43}$	87.30$_{\pm 0.42}$	66.99$_{\pm 0.01}$	82.06$_{\pm 0.02}$
Yan	81.20$_{\pm 0.02}$	80.56$_{\pm 0.04}$	67.55$_{\pm 0.02}$	N/A
PEEB	80.10$_{\pm 0.09}$	N/A	N/A	N/A
V2C	78.59$_{\pm 0.11}$	86.18$_{\pm 0.07}$	63.62$_{\pm 0.09}$	N/A
Hi-CBM	**86.00**$_{\pm 0.04}$	**88.97**$_{\pm 0.18}$	**69.79**$_{\pm 0.23}$	**83.69**$_{\pm 0.30}$

Table 7
Comparison of semantic quality between GPT-4 and MedGPT, evaluated using cosine similarity with expert-generated concepts from the HAM1000 dataset [32]. Due to space constraints, class names in HAM1000 are represented by codes. For the full class names, please refer to [32].

Class Codes	akiec	bcc	bkl	df	nv	mel	vasc
GPT-4	0.8948	0.9232	0.8998	0.9174	0.9087	0.8876	0.8957
MedGPT	0.9426	0.9604	0.9350	0.9624	0.9620	0.9625	0.9322

the feature-based upper bound (**Feat**); even the latest baseline, V2C, achieves only around 78% accuracy. In contrast, Hi-CBM maintains performance nearly on par with **Feat**, reaching about 86% accuracy, demonstrating its strong capability for fine-grained visual reasoning. For HAM10000, most baselines (e.g., Yan, PEEB, and the latest V2C) are not directly applicable or fail to generalize well due to their reliance on generic visual or linguistic concepts. Nevertheless, Hi-CBM remains effective in this medical domain and delivers the best overall performance among all CBMs, highlighting its robustness and adaptability across specialized tasks.

4.4. Information leakage evaluation

To comprehensively assess concept leakage, we adopt three evaluation metrics: **Concept Locality** [37], **Concept Perturbation** [38], and our proposed metric, **Concept Leakage Evaluation** (CLE). Concept Locality and Concept Perturbation are widely used to measure robustness properties such as spatial localization and sensitivity to concept-level perturbations. However, they offer limited insight into the extent to which concept labels contribute to final class predictions. To complement these perspectives, we propose Concept Leakage Evaluation (CLE), a novel metric inspired by prior work [20], which quantifies leakage by progressively removing concepts and measuring the corresponding decline in CBM accuracy. A larger drop indicates stronger reliance on the removed concepts, reflecting greater leakage. Since all three metrics require fine-grained visual annotations, we conduct our evaluation on the CUB-Bird dataset [31], which provides detailed concept-level labels and enables fair comparisons across different evaluation protocols.

4.4.1. Concept locality evaluation

The metric Concept Locality [37] checks whether modifying image regions outside a concept will affect its corresponding probability predicted by the CBM. For this experiment, we set up the evaluation using the CUB-Bird dataset, which provides comprehensive annotations for the concepts of beak, wings, and eyes. The dataset includes 79 images with full concept annotations, making it ideal for evaluating leakage. In our evaluation, every time a CBM predicts one concept c for an image x, we select another different concept c^* from the annotated ones and remove all pixels belonging to c^* from x. We then feed the modified x (where c^* is removed) to the CBM and record the absolute change in c's probability. The less change in the concept c's probability, the less information is leaked. Formally, the Concept Locality for concept c is calculated as:

$$\mathrm{CL}(c) = \frac{1}{N}\sum_{i=1}^{N}\left|p_c(x^{(i)}) - p_c\left(x^{(i)} \setminus c^*\right)\right|, \tag{4}$$

where $p_c(x)$ denotes the predicted probability of concept c for image x, c^* is a different annotated concept whose corresponding pixels are removed from x, and N is the total number of evaluated samples. Lower $\mathrm{CL}(c)$ indicates less locality leakage.

As illustrated in Table 8, Hi-CBM exhibits significantly lower locality leakage compared to other CBMs, with locality values of 0.0104, 0.009, and 0.014 for beak, eye, and wing, respectively.

4.4.2. Concept perturbation evaluation

Concept Perturbation [38] identifies irrelevant (i.e., leaked) information by checking whether the label prediction result remains unchanged when perturbing concept prediction results. If modifying some concepts does not change the predicted label, the modified concepts denote irrelevant information. [38] proposes to find such cases (i.e., altered concept prediction but retained label prediction) via gradient-based optimization. We implemented the Concept Perturbation metric based on Projected Gradient Descent (PGD) [39] and updated its objective to simultaneously minimize label prediction loss (i.e., retaining the label prediction) and maximize the concept prediction loss (i.e., differing the concept prediction). Since the ground truth concepts are unavailable, for each input, we record the maximal concept probability variation that does not change the original label prediction result. The maximal variations are represented in the form of L_2 distance norm and averaged for all inputs. Formally, for input x, the metric is computed as:

$$\mathrm{CP} = \frac{1}{N}\sum_{i=1}^{N}\left\|\mathbf{p}(x^{(i)}) - \mathbf{p}\left(x^{(i)}_{\mathrm{pert}}\right)\right\|_2, \tag{5}$$

where $\mathbf{p}(x)$ is the vector of predicted concept probabilities for x, x_{pert} is the perturbed input obtained via PGD such that the predicted label remains unchanged, and N is the total number of evaluated samples. Lower CP indicates less leakage of irrelevant information.

We set up the pipeline with various configurations for the step size α and the perturbation bound ϵ, and report the L_2 norm values in Table 9, where a lower value indicates less leakage. We use the CUB-Brid dataset, the same dataset used in Section 4.4.1. It is apparent that Hi-CBM's perturbation leakage is significantly lower than other CBMs.

4.4.3. Concept leakage evaluation

For each evaluated CBM, we start by training it with the full concept set. Then, during the inference, we rank the concepts based on their importance and gradually remove top-ranked concepts. We expect that the performance of CBMs which exhibit better resilience to information leakage should drop more quickly when more concepts are removed. Here, we use our constructed concept set for a fair comparison. The intuition behind the CLE evaluation is that, if a concept set is insufficient, the label prediction has to exploit additional information to fulfill the training objective (e.g., achieving high accuracy). In that sense, if a CBM suffers from information leakage, removing concepts that contribute most to the label prediction should *not* notably undermine the CBM performance. Formally, let n be the total number of concepts and $k = \lfloor r \cdot n \rfloor$ be the number of removed top-ranked concepts. Define:

$$\mathrm{CLE}(r) = \frac{1}{N}\sum_{i=1}^{N}\mathbb{1}\left[y^{(i)} = \hat{y}^{(i)}_{\setminus k}\right], \tag{6}$$

where $y^{(i)}$ is the ground-truth label for sample i, $\hat{y}^{(i)}_{\setminus k}$ is the predicted label when the k most important concepts are removed from that sample's concept set, and $\mathbb{1}[\cdot]$ is the indicator function, which equals 1 if the

Table 8
Evaluation of Concept Locality [37] for Hi-CBM on the CUB-Bird dataset. The table presents the locality leakage values for the concepts of beak, eye, and wing, demonstrating that Hi-CBM exhibits significantly lower locality leakage compared to other Concept-Based Models (CBMs).

	Beak	Eye	Wing
LaBo	1.43	1.194	1.35
Yan	1.03	0.902	1.74
Label-Free	0.853	0.824	1.12
Hi-CBM	**0.0104**	**0.009**	**0.014**
PEEB	1.01	0.952	1.11

Table 9
Comparison of L_2 norm values for Concept Perturbation [38] across various configurations of step size α and perturbation bound ϵ on the CUB-Brid dataset. Lower values indicate reduced leakage of irrelevant information. Hi-CBM demonstrates significantly lower perturbation leakage compared to other CBMs.

	$\epsilon = 2/255, \alpha = 1/255$	$\epsilon = 4/255, \alpha = 1/255$	$\epsilon = 8/255, \alpha = 2/255$
LaBo	0.305	0.44	0.55
Yan	0.437	0.767	1.09
Label-Free	0.414	0.64	0.872
Hi-CBM	**0.023**	**0.034**	**0.054**
PEEB	0.397	0.652	0.964

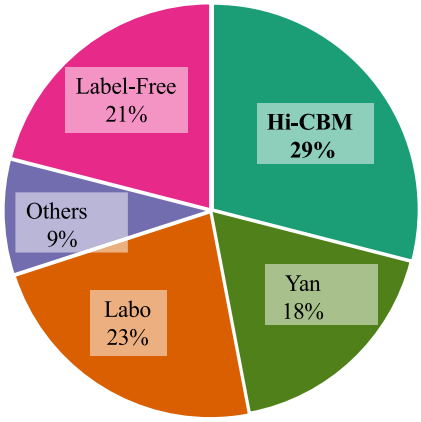


Fig. 4. Results of a human study evaluating the interpretability of concept predictions from Hi-CBM and other CBMs from the CUB-Bird dataset. Hi-CBM was selected as the most interpretable.

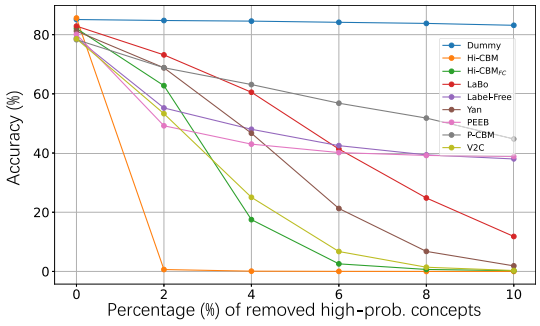


Fig. 5. Results of the Concept Leakage Evaluation on the CUB-Bird dataset, comparing Hi-CBM with seven baseline methods. In this evaluation, a CBM demonstrating higher resilience to information leakage should exhibit a more rapid decline in performance as progressively more concepts are removed. Notably, Hi-CBM shows the most significant decline, indicating its strong capability in mitigating leakage on fine-grained visual recognition tasks.

condition inside is true and 0 otherwise. The CLE curve is then the plot of CLE(r) over different $r \in [0, 1]$.

As shown in Fig. 5, we can see that the dummy model exhibits the lowest resilience to information leakage, as its performance drops slowly when more concepts are removed. Specifically, even when top 10% important concepts are removed, the dummy model still has ~80% accuracy for the CUB-Bird. Fig. 5 also shows that Hi-CBM's performance drops the fastest when more concepts are removed, indicating its highest resilience towards information leakage compared to other CBMs. We attribute this performance gap to our label-supervised concept prediction approach, which effectively eliminates irrelevant concepts, thereby enhancing resilience to information leakage [20]. Furthermore, a cross-comparison between Hi-CBM and Hi-CBM$_{FC}$ reveals that the label prediction facilitated by the intervention matrix also contributes to improved resilience against information leakage, as discussed in Section 3.3. In the next two assessments, Concept Locality and Concept Perturbation, we evaluate LaBo, Yan, Label-Free, and PEEB—identified from CLE results as competitive SOTA methods—against our proposed Hi-CBM to assess its effectiveness in mitigating leakage.

4.5. Interpretability evaluation

This section focuses on the interpretability of Hi-CBM's concept predictions. Specifically, we examine whether Hi-CBM produces predictions that are not only aligned with human perception, but also whether these predictions are robust, accurate, and distinguishable from hallucinated or misleading interpretations often observed in large-scale vision-language models (VLMs). We begin by assessing human perception of interpretability through a carefully designed user study. We then provide qualitative visual analyses to illustrate how Hi-CBM captures meaningful and discriminative concepts across diverse image datasets. Finally, we compare Hi-CBM's interpretability to that of recent VLMs, demonstrating its superior robustness and consistency in generating reliable explanations.

4.5.1. Human-centric evaluation of concept interpretability

To evaluate the interpretability of Hi-CBM's concept predictions, we conduct a large-scale human study on the Amazon Mechanical Turk platform [40]. Our study is formed using 100 questions, and each question consists of one image from the CUB-Bird dataset and concept predictions of Hi-CBM and other CBM baselines listed in Section 4.1. For each question, we ask 11 pH.D. students experienced in computer vision projects to select the most interpretable concept prediction. The interpretability is assessed from three aspects: 1) the relevance to the image content, 2) the completeness of covered image semantics, and 3) the clarity of the described concepts. The most frequently selected concept prediction is deemed as the answer to the question. We also prepare a 20-minute teaching session before the study. Results are shown in Fig. 4. Hi-CBM is selected as the most interpretable in 29 of 100 cases, largely outperforming other CBMs. LABO has the second-best result, presumably due to its large concept set (i.e., 10,000 concepts) compared with other SOTA CBMs. Hi-CBM, in contrast, has only 3236.

4.5.2. Interpretability of concept recognition

Beyond subjective evaluation, we qualitatively analyze how the concept predictions generated by Hi-CBM provide transparent and human-understandable explanations for model decisions on two mainstream benchmark datasets on interpretability: CUB-Bird [31] and Flowers [41]. By leveraging our proposed concept pooling technique, Hi-CBM not only identifies precise intra-class concepts but also makes its decision process interpretable by explicitly attributing predictions to meaningful visual concepts. As illustrated in Fig. 6, for each image, we first present the perceptual concepts (i.e., first-level concepts) identified by Hi-CBM, followed by their most salient descriptive concepts (with the highest probability).

Consider the first row of Fig. 6, where three black-footed albatrosses exhibit visually distinct wing postures. Hi-CBM demonstrates high interpretability by explicitly distinguishing these differences through semantic concept recognition. For instance, the third albatross, with its wings spread, is accurately characterized by the concept "`Slight dihedral shape`" for its wings. Similarly, the second albatross, with folded wings, is described by the adjectival concept "`narrow`" for both "`tail`" and "`wing`". These explicit, fine-grained concept attributions enable users to directly understand the rationale behind Hi-CBM's predictions, enhancing trust and transparency.

For the Flower dataset (last two rows of Fig. 6), although intra-class visual differences are more subtle compared to the CUB dataset, Hi-CBM continues to provide interpretable and meaningful explanations. For example, in the Petunia images (third row), Hi-CBM precisely identifies "`five sepals`" for the "`calyx`" in the third image, and "`Pointed tips`" in the other two images. These results demonstrate that Hi-CBM not only captures subtle visual distinctions but also translates them into clear, human-interpretable concepts. Overall, these case studies highlight the superior interpretability of Hi-CBM, as it consistently provides transparent and semantically rich explanations for its predictions.

4.5.3. Comparative interpretability analysis with VLMs

To further validate the interpretability of concept-based explanations, we compare Hi-CBM with two state-of-the-art Large Vision-Language Models (VLMs): Intern2VL-8B [42] and Qwen2VL-7B [43]. The visual results are presented in Fig. 7. While VLMs are capable of generating vivid and detailed descriptions, their explanations often lack direct visual grounding and may suffer from hallucinations. For instance,

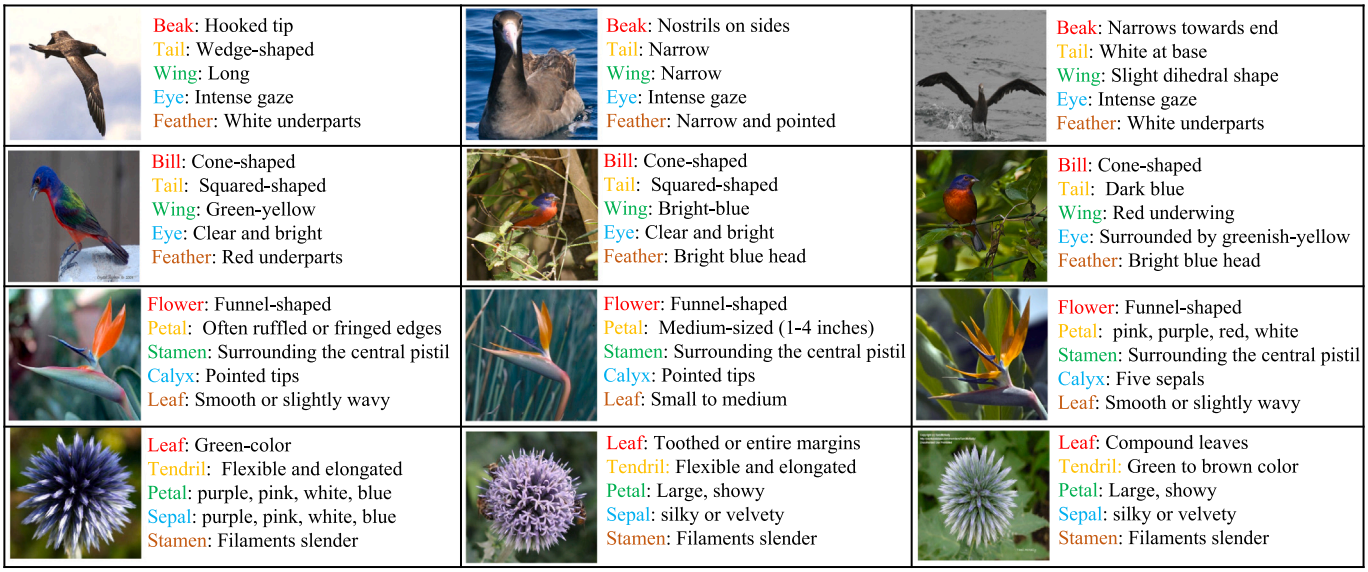


Fig. 6. Case study of Hi-CBM on CUB-Bird and FLOWER datasets.

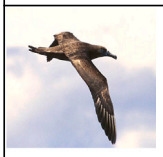
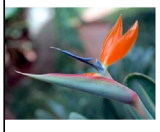
	Hi-CBM	Intern2VL-8B	Qwen2VL-7B
	Beak: Hooked tip Tail: Wedge-shaped Wing: Long Eye: Intense gaze Feather: White underparts	The image shows a bird in flight against a cloudy sky. Its wings are fully extended, displaying an impressive wingspan. The feathers on the wings are primarily dark brown or black, with fine details that reveal the structure and individual feathers. The tail is short and straight, aiding in the bird's aerodynamic form. The bird's legs are tucked close to its body, minimizing air resistance as it soars through the air. The background consists of a light blue sky with scattered white clouds, providing a contrasting backdrop that highlights the bird's dark	The image depicts a large bird, likely a seabird such as an albatross or petrel, soaring through the sky. The bird has a dark brown body with lighter patches on its wings and tail. Its wings are spread wide, showcasing their impressive span and the intricate feather patterns. The background consists of a cloudy sky, providing a natural backdrop that highlights the bird's majestic flight. The overall scene conveys a sense of freedom and grace as the bird glides effortlessly through the air.
	Bill: Cone-shaped Tail: Squared-shaped Wing: Green-yellow Eye: Clear and bright Feather: Red underparts	This image features a vibrant bird perched on a white railing. The bird's plumage showcases a striking array of colors, with a mix of blue, green, red, and yellow. It has a distinctive long beak and is facing towards the left side of the image. The background is blurred, focusing the attention on the bird. The bird's position on the railing and its vivid colors make it the focal point of the image.	The image shows a vibrant and colorful bird perched on a surface. The bird has a striking blue head, a red throat and chest, and greenish-yellow wings. Its tail feathers are dark with hints of red. The background appears to be a wooden fence or wall. The bird's bright colors and unique markings make it stand out prominently against the more muted background.
	Flower: Funnel-shaped Petal: Often ruffled or fringed edges Stamen: Surrounding the central pistil Calyx: Pointed tips Leaf: Smooth or slightly wavy	The image features a vibrant and exotic Bird of Paradise flower. The flower has a striking appearance, with a bright orange and blue beak-like structure resembling the head of a bird. The petals extend gracefully, with hues of blue and green. The background is a soft, blurred green, suggesting lush foliage and a garden setting. The focus is sharply on the flower, highlighting its unique and vivid colors against the muted background. The overall impression is one of natural beauty and tropical elegance.	The image features a vibrant Bird of Paradise flower, characterized by its striking orange and blue petals. The flower is set against a backdrop of lush green foliage, creating a vivid contrast. The petals are elongated and pointed, with the orange ones forming a beak-like structure that encases the blue inner petals. The overall composition highlights the exotic beauty of the flower, emphasizing its unique shape and coloration.
	Leaf: Green-color Tendril: Flexible and elongated Petal: purple, pink, white, blue Sepal: purple, pink, white, blue Stamen: Filaments slender	This image features a close-up of a vibrant blue spherical flower with spiky petals. The intricate details of the petals create a striking, almost geometric pattern. The background is blurred with green hues, accentuating the flower's vivid color and unique texture.	The image features a close-up of a blue globe thistle flower. The flower has numerous small, spiky petals radiating outward from a central point, creating a spherical shape. The petals are a vibrant shade of blue, with some darker and lighter variations. The background is blurred, drawing focus to the intricate details of the flower's texture and color. The overall appearance is striking and visually appealing, highlighting the natural beauty of the plant.

Fig. 7. Case study comparing two VLMs with Hi-CBM on the CUB-Bird and FLOWER datasets.

InternVL describes the bird's legs in the first image as "tucked close to its body, minimizing air resistance as it soars through the air," a statement that cannot be visually verified. Similarly, QwenVL misidentifies the background in the second image as "a wooden fence or wall," despite insufficient visual evidence.

In contrast, Hi-CBM excels in interpretability by providing concise, accurate, and visually grounded explanations. Rather than relying on potentially imaginative narratives, Hi-CBM attributes its predictions to explicit, observable concepts, ensuring that every explanation is both meaningful and verifiable. This approach tends to minimize hallucinations and may enhance user trust since the reasoning for each prediction is made more transparent and grounded in visual evidence. In summary, although VLMs can generate more detailed descriptions, Hi-CBM tends to offer explanations that are more interpretable and accessible to users.

4.6. Intervenability evaluation

To demonstrate Hi-CBM's enhanced intervenability for concept interventions (as detailed in Section 3.2.2), we conduct a local intervention experiment, adhering to the procedure described in [15]. We first collect 300 test inputs from the CUB-Bird dataset that all prior CBMs and Hi-CBM misclassify. Then, we check whether the incorrect label predictions of these inputs can be fixed by removing the top-0.5% (in terms of concept probabilities) concepts predicted by each CBM. In Hi-CBM this can be directly done by setting the corresponding entries in the intervention

Table 10
Results of the intervention evaluation, showing the number of correctly fixed label predictions for each CBM.

	LaBo	Yan	Label-Free	Hi-CBM	Hi-CBM $_{rand}$	P-CBM	PEEB
#correctly fixed predictions	28	24	16	88	9	32	23

matrix as 0 (as described in Section 3.2.2). For all prior CBMs, since they do not provide an intervention mechanism, we manually set the top 0.5% concepts' probabilities as 0. Notice that we also include Hi-CBM with a random binary matrix Hi-CBM $_{rand}$ which behaves the least in this task. We show the number of correct label predictions in Table 10. Our Hi-CBM performs the best, with 88 correct predictions, which is more than double the second-best result of 28 by LaBo.

5. Conclusion

In this paper, we addressed the critical information leakage issue in CBMs, a problem that hinders model interpretability and interventionability by allowing unintended information to influence predictions. To overcome this limitation, we proposed Hi-CBM, a new paradigm for CBMs that leverages structured hierarchical concept sets and explicit label supervision during concept prediction. By utilizing a carefully designed intervention matrix, Hi-CBM significantly improves label prediction performance, concept interpretability, and interventionability. Thorough evaluations across multiple datasets demonstrate that Hi-CBM outperforms existing state-of-the-art CBMs, notably reducing information leakage. Our approach provides a robust and understandable model framework, paving the way for more trustworthy and explainable AI systems.

6. Impact

Hi-CBM is a novel approach to improve the interpretability and intervenability of CBMs. Hence, it can be applied to any existing backbone models, and thus can be easily adopted by practitioners. In light of Hi-CBM's high interpretability, intervenability, and performance, it can be used to improve the transparency and trustworthiness of AI systems in various domains, such as healthcare, finance, and autonomous driving. For example, in autonomous driving, the decision-making process needs to distinguish between genuinely relevant cues and coincidental correlations. A conventional model might incorrectly learn that the presence of pedestrians is associated with a green traffic light, potentially resulting in hazardous decisions. By contrast, Hi-CBM leverages structured interventions to ensure that only conceptually relevant factors contribute to class predictions, thereby mitigating the risk of such critical errors. Similar principles can be applied to healthcare (e.g., avoiding spurious correlations between unrelated patient attributes and diagnoses) and finance (e.g., preventing irrelevant market signals from influencing lending decisions). Therefore, Hi-CBM can be used to facilitate the development of AI systems by providing insights into the model's decision-making process, and thus can be used to improve the robustness and reliability of AI systems.

7. Limitations and future work

Concept annotation remains a primary challenge in developing CBMs. Similar to previous post-hoc concept extraction methods, Hi-CBM leverages LLMs and CLIP to automatically annotate concepts. However, two common limitations persist across all post-hoc approaches: (a) the inability of LLMs to reliably generate relevant concepts, and (b) inaccuracies in similarity scores produced by CLIP. Our empirical results across diverse datasets indicate that limitation (a) is generally not problematic for natural images. Nevertheless, when applying Hi-CBM to domains significantly different from natural images, we recommend generating concepts using domain-specific LLMs. Limitation (b) is effectively mitigated by our proposed intervention matrix and concept pooling mechanism. Specifically, our label prediction step utilizes only the most relevant concepts for each label, and the concept pooling mechanism selects concepts based on the top-k probabilities, significantly reducing the impact of incorrect similarity scores.

Moreover, Hi-CBM may also require further enhancements to realize its full potential. As high-quality CBMs often rely on domain-specific LLMs to perform specialized concept annotation. In some critical domains, such as medical imaging, there are currently no sufficiently mature and openly available domain-specific LLMs that can meet this requirement. In future work, we plan to investigate strategies for developing or adapting domain-specific LLMs tailored for precise concept annotation, potentially through collaborations with domain experts and integration of proprietary datasets where appropriate. By enabling Hi-CBM to leverage such specialized models, we aim to further improve its expressiveness, accuracy, and applicability to safety-critical environments.

CRediT authorship contribution statement

Ao Sun: Writing – original draft, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Qingyue Wang :** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Yuanyuan Yuan:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Pingchuan Ma:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Software, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization. **Shuai Wang:** Writing – review & editing, Writing – original draft, Visualization, Validation, Supervision, Resources, Project administration, Methodology, Investigation, Funding acquisition, Formal analysis, Data curation, Conceptualization.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

Data will be made available on request.

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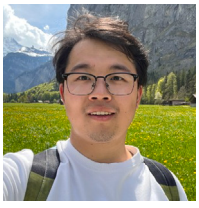
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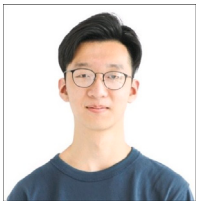
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